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SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2024

DESIGN AND DRAWING OF STEEL STRUCTURES

Time: 3 Hours **Maximum Marks: 50**

Instructions:

i. **PART – A,** Question number 1 and 2 are compulsory. Each question carries 10 marks.

ii. **PART-B,** Answer atleast one full Question. Each question carries 30 marks.

iii. Use of IS 800-2007 and Steel Tables are Permitted.

iv. Missing data may be suitably assumed.

v. Use Drawing sheets to draw the diagrams. All drawing should be done by the aid of Drafter/ Tee Square.

PART - A

1. a. Draw the front and side elevation of two unequal column splicing arrangement with the following data
Lower column : ISHB 300@63 kg/m
Upper column : ISHB 250@51 kg/m **10**
Cover plate: 200X600X20 mm
Distribution plate: 300X200X25 mm

2. a. Draw the Plan, Front Elevation of a Column base with the Following Data:
Column : ISHB 350@67.4kg/m
Base Plate: 700X600X15 mm
Web Cleat: ISA 80X80X10
Flange Cleat: ISA 75X75X10 **10**
Cover Plate: 2No's- 250
Anchor Bolt: 20mm dia- 300mm Long

PART - B

3. Design a simply supported Gantry Girder for the following data,
The girder is electrically operated. Take the yield stress of steel as 250MPa,
Sketch the Details.

i. Span of gantry girder= 8m
ii. Span of Crane Girder = 20m
iii. Capacity of the crane= 230kN **30**
iv. Self weight of the crane excluding crab= 200kN
v. Weight of the crab= 60kN
vi. Wheel base distance= 3.4m
vii. Minimum hook approach= 1.2m

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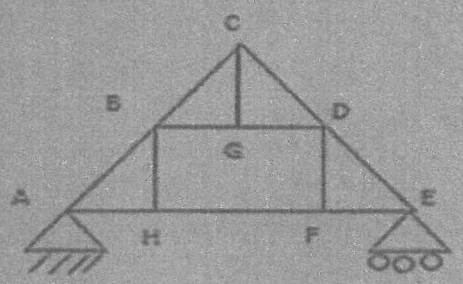
viii. Height of Rail = 70 mm

ix. Self weight of rail= 0.3kN/m

OR

4. Design the Members for the given truss, the size of RC column supporting Truss is 300X300mm. Details of Members are Given Below, Assume Fe250 Steel, M18 G 4.6 Bolts and reactions R_A and R_E as 60kN. Also Design Anchor bolt for uplift of 15kN at each support. **30**



Member	Force (KN)	Length (m)
AB, BC	-150	5
CD,DE	-100	5
AH,FE	95	3.5
BH,DF	105	3.5
BG,GD	-80	2.5
CG	90	2.5
HF	120	4.5